MUESLI EDA PROJECT · DELIVERY KPIS

How long does a muesli order really take?

Warehouse and logistics company assumptions VS reality

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THE BRIEF

Two partners describe the process. We test both.

Warehouse manager

- Processing takes about 2 days
- Warehouse works Mon–Fri; weekend orders wait for Monday
- Trucks leave Mon, Wed and Fri
- Standard: next truck after ready · Express: same day

Logistics company (3PL)

- Average delivery time of 3 days to all locations
- Goods move on weekends
- Deliveries to customers on weekdays only

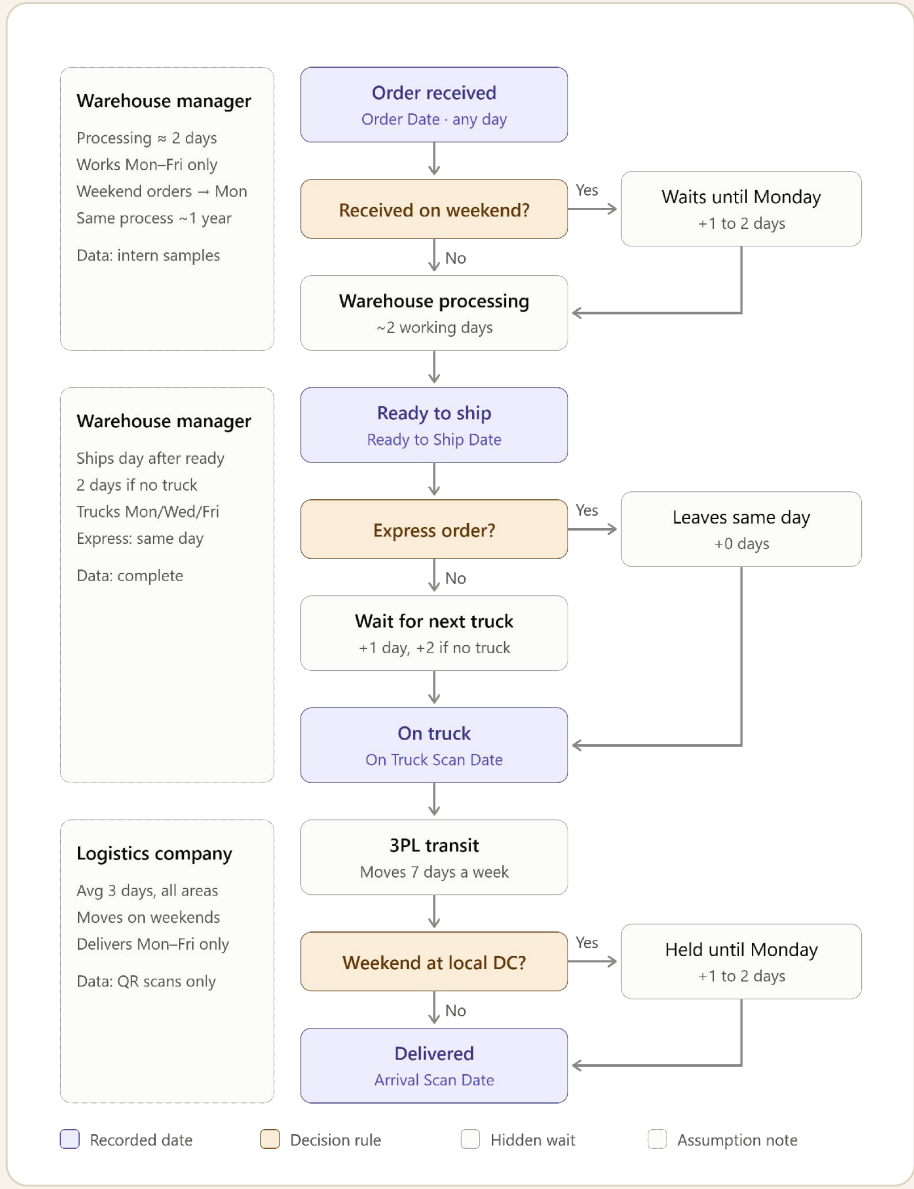
Our job: turn each claim into a measurable KPI and check it against the data.

THE ORDER PROCESS

Four recorded dates, three hidden waits

Each recorded date marks the end of a stage. The time between two dates is the metric for that stage.

Stage	From → to	Assumed (days)
Processing	order → ready	~2 working days
Wait for truck	ready → truck	0 Express 1–2 Standard
Transit	truck → arrival	3
Order → truck	order → truck	~3



THE DATA

Challenge: limited data available in some stages

ORDERS

5,009

orders 2017–2020 with customer, region, channel, sales

ORDER PROCESS

3,002

orders 2019–2020 with order date + truck date for every one

INTERN STUDY

204

orders with a ready-to-ship date (7% of orders)

QR CAMPAIGN

333

orders with an arrival scan date (11% of orders)

Cleaning: 86 duplicated rows in the intern sheet removed · intern pickup date always equals the truck scan date · 1 order with two truck dates (latest kept)

COMBINING THE SHEETS

One row per order, then left merges

groupby first

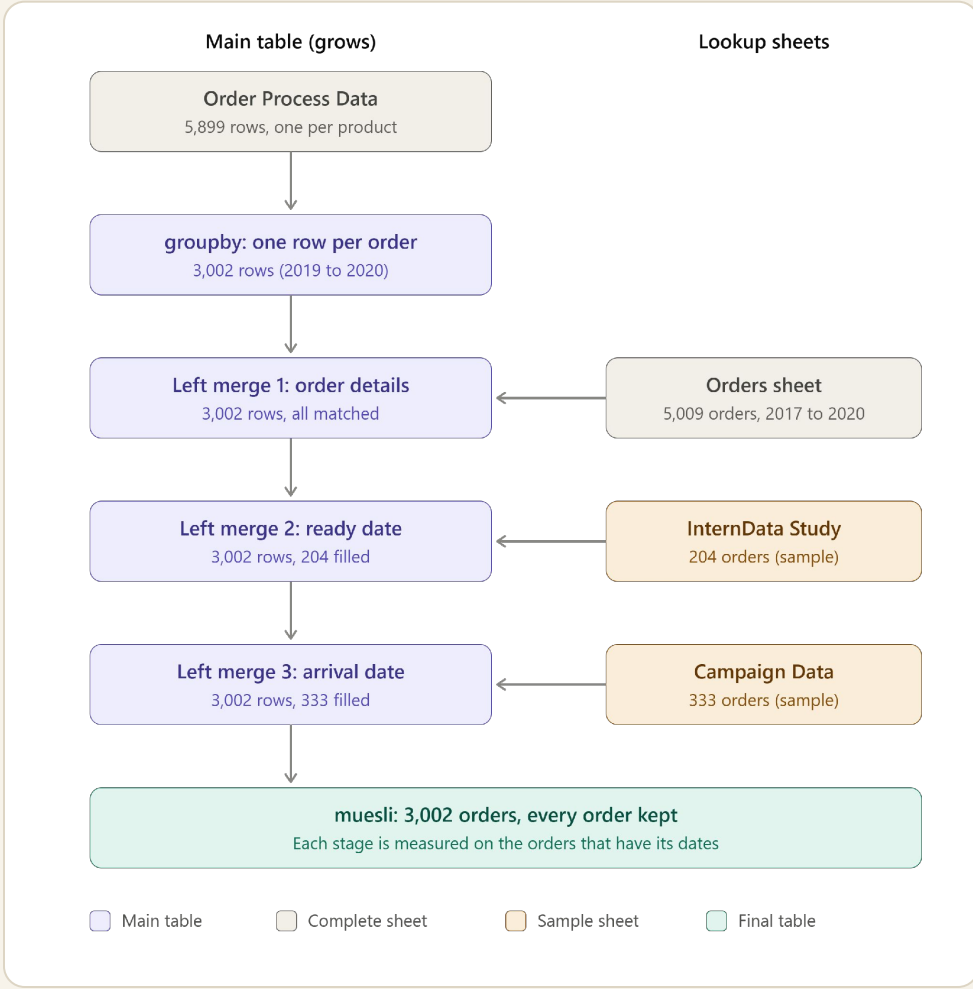
5,899 product rows → 3,002 orders. Without it, orders with many products count several times (418 “ready dates” instead of 204).

Left keeps every order

Samples fill in where they have dates; each stage is measured on the orders that have both of its dates.

Why not inner or outer?

Inner leaves only 22 orders. Outer adds 2,007 older orders with no process dates.



HOW WE BUILT THE METRICS

Dates become durations in pandas

```
# 1. one row per order
muesli = products.groupby('order_id').agg(
  order_date=('order_date', 'first'),
  on_truck_scan_date=('on_truck_scan_date', 'max'))
# 2. left merges keep all 3,002 orders
muesli = muesli.merge(intern, on='order_id', how='left')
muesli = muesli.merge(campaign, on='order_id', how='left')
# 3. calendar days and working days
muesli['transit_days'] = (muesli.arrival_scan_date
  - muesli.on_truck_scan_date).dt.days
muesli['processing_bdays'] = np.busday_count(order, ready)
```

Working days

The warehouse is closed at weekends, so we count Mon–Fri too: Sat → Wed is 4 calendar days but 2 working days.

Expected dates

We replay the manager's rules for each order and compare the expected truck date with the real one.

THE KPIS

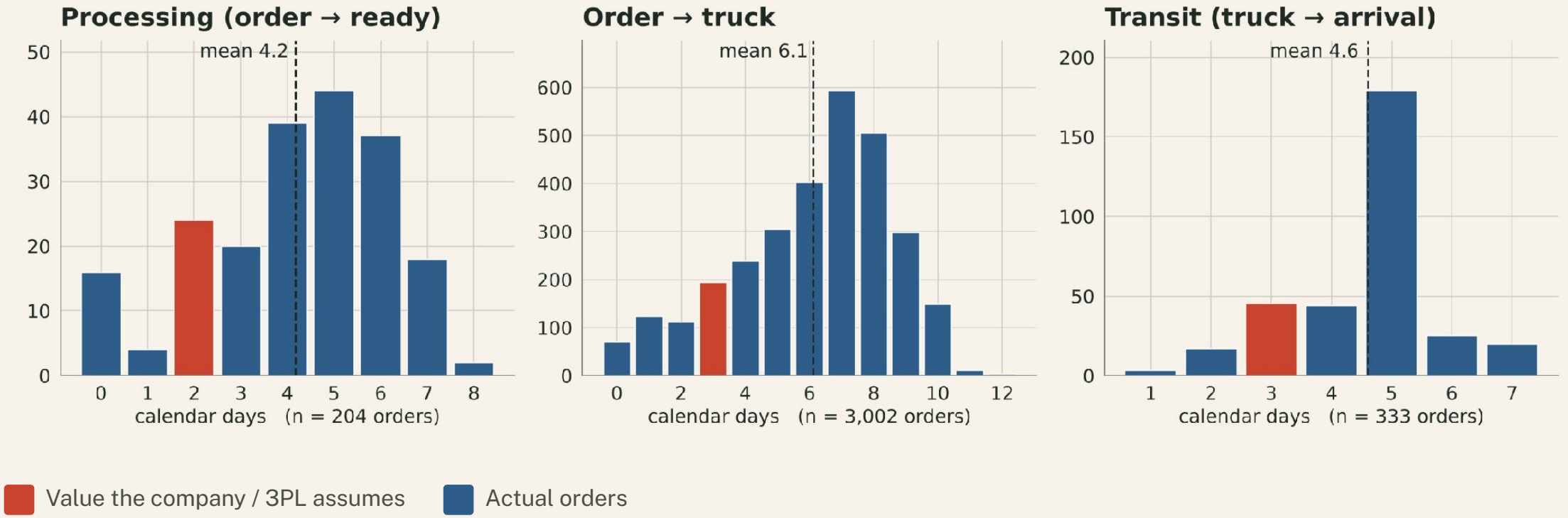
Six metrics, one per question the business asks

KPI	Calculation	Orders	Why it matters
Order → truck	truck − order	3,002	Internal lead time; the only complete one
On time to truck	truck ≤ expected truck	3,002	Is the warehouse following its own rules?
Processing time	ready − order	204	Warehouse capacity
Truck wait	truck − ready	204	Cost of the Mon/Wed/Fri schedule
Transit time	arrival − truck	333	Is the 3PL keeping its promise?
Total delivery time	arrival − order	333	What the customer experiences

For each: mean, median, 95th percentile and variation (coefficient of variation = std ÷ mean).

TIME PER STAGE

Most orders take longer than anyone assumed



MATCH OR MISMATCH?

The schedule rules hold. The time estimates don’t.

Assumption	Actual	Match/Mismatch?
Processing ≈ 2 working days	mean 2.8; only 22% exactly 2, 61% longer	Mismatch
Transit averages 3 days	mean 4.6; 80% take longer than 3	Mismatch
Trucks Mon / Wed / Fri only	4% scanned on a Tue or Thu	Mostly match
Standard: next truck after ready	rule followed for 90% of orders	Mostly match
Express: same day as ready	rule followed for 94% of orders	Mostly match
Deliveries on weekdays only	0% weekend arrivals	Match

HEADLINE KPI · ALL 3,002 ORDERS

57%

of orders reach the truck later than the warehouse's own process says they should

Standard orders

2.4 days late on average

Express orders

0.6 days early on average

IS EXPRESS WORTH IT?

Express arrives five days sooner, mostly thanks to faster processing



Average order → arrival, campaign sample (75 Express, 258 Standard orders)

Average days	Express	Standard	Difference
Processing (order → ready)	2.0	4.9	−2.9
Wait for truck	0.4	2.0	−1.6
Transit (3PL)	4.5	4.6	−0.1

HOW RELIABLE IS EACH STEP?

Warehouse processing is slow and hard to predict

Step (days)	Promised	Average	Usual range	95% within
Warehouse processing	2	4.2	3–6	7
Waiting for the truck	1–2	1.6	1–2	3
Delivery (transit)	3	4.6	4–5	7
Order to truck	*about 3*	*6.1*	*5–8*	*10*
Order to customer	*about 6*	*10.8*	*9–13*	*15*

What this means

- Processing takes twice as long as promised, and it varies the most: anywhere from 3 to 6 days
- The truck wait follows the timetable: 91% of orders catch the truck they should
- Delivery is slower than promised, but steady: almost always 4 to 5 days
- In total, an order reaches the customer in about 11 days, not 6

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DATA QUALITY

Data to treat with care

1

Ready to Ship: 204 orders

Recorded by interns in autumn 2019 and November–December 2020, which is peak season, so times may run high.

2

Arrival: 333 orders

From a QR-code prize campaign, May 2019 – May 2020. Assumes customers scanned on the day of arrival.

3

Only 22 orders have both dates

The full journey can't be measured for most orders, so the next step is a simulation.

4

Edge cases are real, not errors

16 orders were ready the same day (all Express); 4% left on a Tuesday or Thursday, when no truck runs.

RECOMMENDATIONS

Four steps to a healthier delivery process

1

Track order → truck every week

Data already exists for every order. Today only 43% reach the truck on time (Express 82%, Standard 32%).

2

Find out why Standard waits

Standard processing takes 4.9 days, Express only 2.0 (intern sample). Check whether Express jumps the queue.

3

Agree a calendar-day transit target

Transit is 4.6 calendar days (about 3 working days). Customers count calendar days, so set the target in those.

4

Scan every stage

Record ready-to-ship and arrival dates for all orders instead of relying on interns and promotions.

TAKEAWAYS

An order takes **about 11 days** to reach the customer, not the 6 the partners describe.

The rules are followed; the **time estimates are wrong**.

Express pays off: it is processed 2.4 times faster in the warehouse, not just shipped sooner.

Next: simulate delivery times, and compare daily truck pickups with all-Express

Questions?